

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Explain construction and working of a fractionating column with neat sketch.
- Q.24 Discuss the working of Swenson-Walker crystallizer with the help of diagram.
- Q.25 Describe plug flow reactor with diagram.

No. of Printed Pages : 4

220653

Roll No.

5th Semester/Chemical (Pulp & Paper) Subject: Chemical Engineering Operations

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 What is the principle of distillation?
- a) Adsorption
 - b) Absorption
 - c) Difference in boiling points
 - d) Filtration
- Q.2 Crystallization is a process of:
- a) Heating
 - b) Boiling
 - c) Solid formation
 - d) Melting
- Q.3 Which of the following is not a membrane process?
- a) Dialysis
 - b) Decantation
 - c) Microfiltration
 - d) Ultrafiltration

Q.4 Ideal gas law is given by:

- a) $PV = nRT$ b) $E = mc^2$
c) $F = ma$ d) $P = h\rho g$

Q.5 Molecularity refers to:

- a) Product moles
b) Reactant moles involved in one step
c) Number of catalysts
d) Energy loss

Q.6 PFR stands for:

- a) Perfect Fluid Reactor
b) Plug flow Reactor
c) Parallel Flow Reactor
d) Pressure Flow Reactor

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Define relative volatility.

Q.8 Name one type of crystallizer.

Q.9 What is internal energy?

Q.10 Name different types of membranes.

Q.11 Name any one type of reactor.

Q.12 _____ method is used to find no of theoretical stages.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Explain the concept of azeotropic distillation.

Q.14 Discuss the mechanism of crystallization.

Q.15 Explain reverse osmosis with a diagram.

Q.16 Explain closed and open system with one example.

Q.17 Differentiate between batch and continuous reactors.

Q.18 Describe the procedure of McCabe-Thiele method with a neat diagram.

Q.19 Explain the principle of membrane separation.

Q.20 Explain various thermodynamic properties.

Q.21 Describe different types of distillation.

Q.22 Explain order and molecularity of reactions.